# Statistical learning

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# Content

- Introduction
- How we learn rules
- Descriptive vs Inferential statistics

# Introduction

- Statistical analysis saw the development of two branches in the 18th century-

➢ Bayesian Statistics(Believer):Uses **Prior Belief + New Data** to reach a conclusion.

➢ Classical Statistics(Observer):Uses **Only the Current Data**.

- Both deal with probability, but in different ways.

# How we learn rules

- **Deductive (Top-Down):** Start with a **Rule**, apply it to **Data**.
- Logic: “All humans need water” →"John is human" → “John needs water”
- **Inductive (Botttom-up):** Start with a **Rule**, apply it to **Data**.
- *The "Baby" Example:* Touching a hot stove + hot soup + hot iron = "Don't touch hot things!"

# Descriptive vs. Inferential Statistics

| **Type** | **Purpose** | **Example** |
|---|---|---|
| **Descriptive** | Summarizes the **Past**. | A fitness app showing you walked **10,000 steps** yesterday. |
| **Inferential** | Predicts the **Future/Whole**. | Asking **1,000 voters** their opinion to predict how **30 million** will vote. |

# Finding signals in the Mess

- **What is "Noise"?** Random errors or irrelevant data that distract from the pattern.
- **Signal:** Your friend’s voice (the data you want).
- **Noise:** The loud music and crowd shouting (the distraction).
- **Goal:** Machine Learning acts like "Noise-Cancelling Headphones"—it filters out the music to hear the message.

# Explaining noise in terms of data

| House Size (sq.ft) | Actual Price (₹ Lakhs) |
|---|---|
| 1000 | 50 |
| 1200 | 60 |
| 1500 | 75 |
| 1800 | 90 |
| 2000 | 500 |

- First four rows show a clear pattern:
  Bigger house → Higher price 📈
- But last row:
  2000 sq.ft → ₹500 Lakhs

This value is unusually high and may be:

- Data entry error
- Luxury villa in different location
- Typing mistake
- This abnormal value is called **Noise (Outlier)**.

# Hard vs. Soft Modelling

- **Hard Computing (Analytical):** Following a strict, 100% exact recipe
- **Rule:** If A + B = C then C is always same
- **Example :** When you type 567×432, the computer uses a "hard" mathematical rule. There is no guesswork; the answer is either right or wrong.
- **Soft modelling:** Learning patterns to make a "best guess.“
- **Rule:** Based on thousands of examples, this is probably the answer.
- **Example: : Netflix Recommendations.** There is no "hard" rule that says if you liked *Stranger Things*, you will definitely like *Wednesday*.

# Dealing with messy data

- Data is rarely perfect. We have to manage three main problems:

1. **Noise:** Irrelevant "jitter" or extra info that hides the truth.
2. **Missing Values:** Gaps where data wasn't recorded.
3. **Inconsistencies:** Data that contradicts itself.

| Employee ID | Age | Experience (Years) | Department | Salary (₹ Lakhs) |
|---|---|---|---|---|
| E101 | 25 | 2 | Sales | 4.5 |
| E102 | 30 | 5 | Marketing | 6.8 |
| E103 | 29 | 4 | Sales | **150.0** |
| E104 | **–** | 3 | HR | 5.2 |
| E105 | 32 | **12** | Intern | 3.0 |
| E106 | 28 | 4 | Sales | 6.0 |

# Sample Space & Sample Points

- **The Sample Space (D):** This is the "world" of all possible outcomes. In Machine Learning, your entire dataset (the **Data Matrix**) is your Sample Space.
- **Example:** If you are analysing a deck of cards, the Sample Space is all 52 cards.
- **Sample Points (x):**This is a single "event" or a single row in your data. It is one specific instance from the Sample Space.
- **Example:** Drawing the 7 of Hearts is a "sample point." In a medical study, one single patient’s records would be a sample point

# Random Variables

- We call the characteristics of our data "Random Variables" because we don't know the value until we actually look at a specific example.
- These are the **Columns** in your table (Attributes).
- **The "House Hunting" Example:** If you are looking for a house, the **Random Variables** are:
- Number of Bedrooms
- Square Footage
- Price

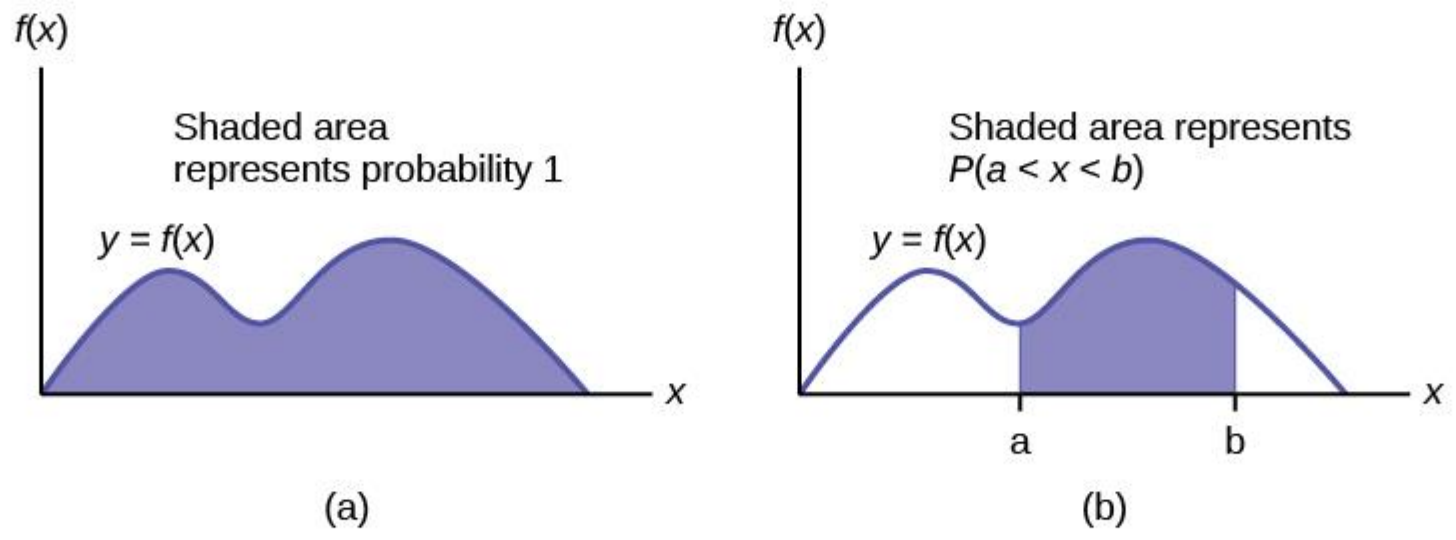
- **The "Zero Probability" Problem**
- Since there are infinite possible values: The probability of getting **exactly one specific value** is ZERO.
- Instead of asking: What is probability of exactly 20 minutes?
- We ask:
- What is probability that commute time is between 18 and 22 minutes?
- For continuous variables, we always find probability for a **range**.
- Instead of looking at a single point, we look at the probability of a value falling within a **range** (between $V_{1x}$ and $V_{2x}$)

$$P(v_{1x} \le x < v_{2x}) = \int_{v_{1x}}^{v_{2x}} p(x)\,dx$$

$$p(x) \ge 0, \text{ and } \int_{-\infty}^{\infty} p(x)\,dx = 1 \quad (3.3)$$

# What is PDF

- A **Probability Density Function (PDF)** is:
- A smooth curve
- That shows how values are distributed
- Probability is the **area under the curve**
  It looks like a smooth wave (not spikes like PMF).



# How do we calculate Probability

- To find probability between two values:
- We calculate the **area under the curve** between those two points.
- Mathematically, we use **integration**.
- Simple meaning:
- Probability = Area under curve between $V_1$ and $V_2$

# Rules of PDF

- **Rule 1: Density must be zero or positive**
- $p(x) \geq 0$
- You cannot have negative density.
- **Rule 2: Total area under curve must equal 1**
- The entire area under the curve = 1 (which means 100%)
- If total area $\neq 1 \rightarrow$ it is not a valid PDF.

- **Very Small Interval Idea (Δx Concept)**
- If we take a very small range:
- $x_0$ to $x_0 + \Delta x$
- Then,
- Probability ≈ Density at that point × Width of range
- In simple words:
- Small Probability ≈ Height × Width
- Just like area of a rectangle.

$$P(x_0 \leq x < x_0 + \Delta x) \approx p(x = x_0)\Delta x$$

- **Data Scientist perspective:**
- From a data point of view:
- Density means:
- How frequently values occur in a very tiny interval.
- If:
- N = total observations
  ΔN = number of values in small interval
- Then:
- Density = (Relative frequency) ÷ (Width of interval)
- As interval becomes extremely small → it becomes PDF.

$$p(x) = \lim_{N \to \infty, \Delta x \to 0} \left[ \frac{\Delta N / N}{\Delta x} \right]$$

# Descriptive Measures of Probability Distributions

- One of the most important descriptive values of a random variable is the point around which distribution is centered, known as measure of **central tendency.**
- The expected (mean or average) value is the best measure for central tendency.
- Expected value is just the **average** of a random situation.

- **Example 1: Dice Roll (Discrete Case)**
- Suppose you roll a fair dice.
- Possible outcomes:
  1, 2, 3, 4, 5, 6
  Each has probability = 1/6
- Expected value:
- $E[X] = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6)$
- $= 3.5$
- You will **never actually get 3.5**
  But if you roll many times, the average will approach 3.5.
- **For Discrete Data (Countable steps, like a dice roll):** You use simple addition. You multiply each outcome by its probability and sum them up.
- E [x]=μ $= \sum_{x \in S} xp(x)$
- where P(x) is probability mass function

- **Example 2: Rainfall (Continuous Case)**
- Suppose rainfall can be any value like:
- 12.2 mm, 12.25 mm, 12.251 mm…
- Here values are continuous.
- So instead of adding, we use **integration**.
- But concept is same:
- Multiply each value by how likely it is
  Add everything together
- That gives the average rainfall.
- **For Continuous Data (Smooth curves):** You use calculus (integration). You multiply every possible value (x) by its likelihood (p(x)) and add them all up.
- $$E\,[x] = \mu = \int_{-\infty}^{\infty} x p(x)\,dx$$
- where p(x) is the probability density function.

# Numerical

- A data scientist is evaluating a simple "Recommender System" for a streaming app. The random variable  x represents the number of stars (1 to 4) a user is likely to give a specific genre based on historical data.

| x | P(x) |
|---|---|
| 1 | 0.1 |
| 2 | 0.3 |
| 3 | 0.4 |
| 4 | 0.2 |

Questions: Verify that this is valid PMF

Calculate Expected Value E[x]

(Average Rating)

Find Variance and Standard deviations

- A food delivery company tracks number of orders received per hour. The following data was collected:

| Orders per hour (x) | Frequency |
| --- | --- |
| 0 | 5 |
| 1 | 10 |
| 2 | 20 |
| 3 | 10 |
| 4 | 5 |

- **Questions:**
- Construct the probability mass function P(x).
- Verify that it is a valid PMF.
- Find the expected number of orders per hour.
- Find variance and standard deviation.

# Important properties

- **1. Constants stay the same:** The "average" of a constant number (like 5) is always 5.
  E [C] = C; C = constant
- **Example:** If you always get ₹500 salary bonus: $E[C] = C$
- Average of 500 is 500.
- **2. Scaling:** If you multiply all your data by 10, your average also multiplies by 10.
- E [Cx] = C E [x]
- **Example**: If daily salary is ₹1000
  Now company doubles it → ₹2000
- New average also doubles.

- 3. **Additivity:** The average of a bunch of things added together is the same as adding their individual averages.
- E$[\sum_{j=1}^{n} x_j]$= $\sum_{j=1}^{n} E[x_j]$
- 4. **Joint Variables:** When dealing with two variables at once (like height and weight), you use their "joint" probability to find the average of their product.
- If there are two random variables x1 and x2 defined on a sample space, then
- $\mathbf{E}\ [\boldsymbol{x_1}\ \boldsymbol{x_2}\ ] = \iint_{-\infty}^{\infty} \boldsymbol{x_1}\ \boldsymbol{x_2}\mathbf{p}(\boldsymbol{x_1}\ \boldsymbol{x_2})\ \mathbf{d}\boldsymbol{x_1}\ \ \boldsymbol{dx_2}$
- **Why Do We Need Joint Probability?**
- Because now two things are happening together.
- If: $X_1$ =Height, $X_2$ =Weight
- Height and weight are related.
  So we must look at their **joint behaviour** — how they occur together

**5. Functions:** If you want the average of $x^2$ you just plug $x^2$ into the formula where x used to be.

- E [f(x)] = μ= $\int_{-\infty}^{\infty} f(x)p(x)\mathrm{dx}$

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- **Vectors:** If your data is a list of numbers (a vector) rather than a single value, the math works exactly the same way—you just do it for the whole list at once.
- For a vector random variable x,
- $$\mathbf{E\ [x]} = \boldsymbol{\mu} = \int_{-\infty}^{\infty} \boldsymbol{x p(x)}\,\mathbf{dx}$$
- In discrete case,
- $$\mathbf{E\ [x]} = \mu = \sum_{x \in S} \boldsymbol{x p(x)}$$
- 

# Numerical

- A multi-specialty hospital has deployed a **Machine Learning-based patient triage system**. The ML model predicts patient severity and dynamically allocates doctors. After analysing historical data, the **data science team** observes that the waiting time (in minutes) for patients follows a probability density function:
- $p(x) = kx \ for \ 0 \leq x \leq 10$

1. Find Value of k
2. Find probability patient waits between 2 and 5 minutes
3. Find Expected waiting time

# Numerical

- An autonomous vehicle uses a machine learning–based distance estimation model to predict the distance $X$(in meters) to nearby objects. The predicted distance follows a continuous probability density function:
- $p(x) = 0.1 \, for \, 0 \leq x \leq 10$
- Assuming the model output follows this distribution, compute the probability that an object is detected between **2 and 5 meters** from the vehicle.

# Variance

- The expected value (mean) describes the central tendency of a random variable.
- We also need to measure the spread of values around the mean.
- Variance is desired to know the ‘spread’ of the random variable about the mean.
- It is denoted as Var[X] or $\sigma^2$

# Variance (Continuous Case)

- Var[x]= $\boldsymbol{\sigma}^2$ = E $[(x - \mu)]^2 = \int_{-\infty}^{\infty} (\boldsymbol{x} - \boldsymbol{\mu})]^2 \boldsymbol{p}(\boldsymbol{x})\mathbf{dx}$
- It is the weighted average of squared deviations
- from the mean.

# Variance (Discrete Case)

- $Var[X] = \sigma^2 = E[(X - \mu)^2]$

- Summation Form:
- $Var[X] = \Sigma (x - \mu)^2 P(x), \; x \in S$

- It represents weighted squared deviations
- from the mean.

# Derivation of Variance Formula

- $Var[X] = E[(X - E[X])^2]$

- $= E[X^2 - 2X\,E[X] + (E[X])^2]$

- Using linearity of expectation:
- $= E[X^2] - (E[X])^2$

- Final Formula:
- $Var[X] = E[X^2] - (E[X])^2$
- This form is often easier to compute
- in practical problems.